

Salah Boubnider University

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Faculty of Medicine of Constantine

Cytology Courses

"The Cytosolic Compartment"

I. Definitions:

I.1. Cytoplasm: The intracellular environment bounded by the plasma membrane. It is a heterogeneous phase containing organelles and cytoskeletal structures suspended within the cytosol.

I.2. Cytosol: Occupying approximately 54% of the cell volume, the cytosol is the liquid, hyaline, and homogeneous fraction in which organelles are immersed. Technically, it corresponds to the supernatant remaining after the sedimentation of all organelles during differential ultracentrifugation.

II. Ultrastructure and Chemical Composition of the Cytosol:

II.1. Water: Accounts for approximately 85% of the total cytosolic volume.

II.2. Small Molecules: Including ions (Na^+ , K^+ , Ca^{++} , ...etc.) and dissolved gases (O_2 and CO_2).

II.3. Intermediate-sized Molecules: Such as carbohydrates, lipids, amino acids, nucleotides, and various metabolites resulting from cellular metabolism.

II.4. Macromolecules: Such as polysaccharides, nucleic acids, glycoproteins, and proteins (including cytoskeletal proteins and enzymes). This also includes ribosomal subunits, whether dissociated or assembled into functional complexes.

III. Properties of the Cytosol: The cytosol exhibits several key physicochemical properties:

III.1. pH: The cytosolic pH ranges between 7 and 7.4. This neutral pH provides an optimal environment for the function of cytosolic enzymes and the stability of proteins.

III.2. Viscosity: The high protein density of the cytosol results in significant variations in its viscosity (consistency). These variations are localized and dynamic, allowing the cytosol to change rapidly from Sol-to-Gel state and *vice versa*, depending on the cell's physiological state (e.g., protein synthesis, degradation, secretion, division, or motility).

a. Viscous State (Gel): Occurs when cytoskeletal proteins assemble (polymerization). Examples: Cell division and plasma membrane remodeling (formation of lamellipodia, filopodia, and pseudopodia).

b. Fluid State (Sol): Occurs when cytoskeletal proteins disassemble (depolymerization). Examples: Vesicular secretion, metabolic synthesis, and degradation.

III.3. Motility: Two types of movement can be observed:

a. Unstructured movements: Characterized by the circular flow of intracellular fluid (**cytoplasmic streaming** or **cyclosis**), occurring without the direct involvement of cytoskeletal proteins.

b. Structured movements: Movements mediated by cytoskeletal proteins, such as **cytokinesis**, chromosome migration, and amoeboid movement.

IV. Primary Functions of the Cytosol: The cytosol serves as the site for numerous cellular processes:

IV.1. Metabolic Hub: Many metabolic pathways begin in the cytosol and continue in other compartments following metabolite **targeting**. Metabolism encompasses both anabolism and catabolism. *Examples:* Glycolysis, the pentose phosphate pathway, and glycogen metabolism.

IV.2. Protein Synthesis: The synthesis of all proteins begins in the cytosol, whether they are destined to remain there (approximately half) or be **translocated** into the lumen of the RER (Rough Endoplasmic Reticulum). The cytosol is the synthesis site for: all its own proteins (enzymes, cytoskeletal components), nuclear proteins, peroxisomal proteins, most mitochondrial proteins, G-proteins, and extrinsic proteins on the cytosolic face of the plasma membrane.

IV.3. Protein Degradation: Through specialized complexes called **proteasomes**, the cytosol acts as the site for the proteolysis (degradation) of various protein types.

IV.4. Signal Transduction: Various cytosolic molecules participate in signaling pathways by generating **second messengers** such as cAMP and IP3.

IV.5. Storage Site: Sugars and lipids are stored in polymerized or inclusion forms within the cytosol (e.g., glycogen and **lipid droplets**).

Glycogen: Found in most cells except neurons; it is highly abundant in hepatocytes and muscle cells.

Lipid Inclusions (Lipid Droplets): Abundant in cells specialized in storing fatty acids and cholesterol, such as adipocytes, hepatocytes, and steroid-hormone-secreting cells.
